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Effect of the Productivity of Grass Ecosystems on the Underlying-Surface Albedo

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Abstract—The dependence of the relative leaf area on the productivity of grass cover is analyzed on the basis of experimental data. It is established that this dependence is not a one-to-one correspondence. It is shown that the integral albedo of the underlying surface covered with vegetation can be determined as a function of the values of its overground phytomass. The proposed semiempirical model allows one to calculate the albedo of grass cover in relation to the values of its productivity. The threshold values of the grass albedo and phytomass at which the relation between these quantities becomes ambiguous are determined.

INTRODUCTION

The optical characteristics of the underlying land surface, which are mainly determined by the character and properties of vegetation, are of importance in studying climate changes and in solving the problems of atmospheric radiation power. One of the most important optical characteristics of the underlying surface is the reflectivity of plant communities (phytocenoses), which is determined from their albedo values. This characteristic is the sum of reflections from the grass-cover elements (leaves, stalks, etc.) and the soil, whose contribution to the reflections depends on vegetation density. The albedo value is affected by the indices of the productivity of phytocenoses, in particular, their overground phytomass and relative leaf area, which is also called the leaf area index (LAI). The former of these parameters is determined by the amount of dry organic matter of vegetation (expressed in mass units) per unit area; and the latter shows how many times the area of leaves exceeds the area of the base of a vertical column within which these leaves are located.

Albedo is one of the most important indices of the radiation regime of a specific territory and is used as a parameter in global climate models, in models of water–heat exchange between the underlying surface and the atmosphere, and in models of a production process. It is also used in diagnosing the state of grass cover with remote-sensing methods. In [1], it is assumed that the study and refinement of the space–time variability of the integral characteristics of the radiation–heat balance of the Earth–atmosphere system as a whole and albedo in particular will allow one to refine the contribution made by variations in the physical properties of the underlying surface in estimating climate changes in the present and future.

The presence of the physical relations between the state of grass cover and the intensity of energy fluxes in the atmosphere–underlying surface system allows one to assume that each type of vegetation corresponds to a certain albedo value [2]. In some models of climate and heat exchange between the underlying surface and the atmosphere (see, for example, [3, 4]), this hypothesis is used in modeling a climate response to variations in the underlying surface albedo due to both natural and anthropogenic factors [5, 6]. However, the models and methods based on this hypothesis do not allow one to study albedo dynamics under changes in the state of the underlying surface, for example, under changes in the biometric parameters of the plant communities that form both natural grass and agricultural ecosystems. It should be noted that the difference in the albedo values of some vegetation types is small, which does not enable an unambiguous determination of the type of vegetation from the value of its albedo.

Among the factors affecting the energy balance of the climate system, both spatial and temporal inhomogeneities of the underlying surface are of considerable importance. In particular, variations in the reflected solar radiation flux due to the patchy structure and phenological state of grass cover are significant for the energy–mass exchange between the atmosphere and land surface [7, 8]. Therefore, it is of interest to determine the relation between the reflective and structural characteristics of grass cover. The interrelation between the energy dissipated by a concrete ground ecosystem and seasonal vegetation dynamics is analyzed on the basis of a thermodynamic approach in [9]. Using a large body of empirical data for five ecosystems, the authors of [9] showed that the solar energy absorbed by all the components of the underlying surface of each ecosystem reaches its maximum at the moment when the productivity of grass cover is maximum. In [10], on the basis of the generalized results of both the KUREX 88 and 91 experiments, one of the methods of taking into account the patchy structure of the underlying surface is proposed for a river-basin scale, and the dependence of the albedo of grain crops on their phytomass and density is analyzed for the territory under consideration. In [10], the characteristic (of the region under study) values of the structural parameters and albedo of the underlying surface, for which their conjugation and ordering in time exist, are determined. The authors of [11] analyzed the dependence of the albedo of grain crops on the LAI on the basis of a large body of experimental data. The dependence of the albedo of winter and spring crops on the stage of their development is analyzed in [12]. In this work, on the basis of experimental data, an attempt is made to reveal the relation between the albedo and phytometric parameters of vegetation and to analyze the interrelation between the values of the albedo and phytomass of plant communities during their vegetation period.

DESCRIPTION OF THE MODEL

According to a number of works [7, 13, 14, and others] in which the dependence of grass-cover productivity on solar radiation has been studied, photosynthetically active radiation (PAR) absorbed by phytocenosis at a given level can be characterized by the exponential law

$$F(L) = QK\exp(-KL), \tag{1}$$

where Q is the density of the total PAR flux incident on phytocenosis (W/m^2), L is the LAI for the layer above the given level (m^2/m^2), and K is the extinction coefficient, which depends mainly on the optical properties of leaves and their inclination and mutual arrangement. In [15, 16], to calculate the absorbed PAR for an ideal phytocenosis, the following formula is obtained:

$$F(L) = \frac{Q(1-A_f)(1-a_T)}{L_0}, \tag{2}$$

where A_f is the albedo of grass cover within the PAR spectral region, a_T is the function of light-flux transmission by grass cover within the PAR region, and L_0 is the LAI (m^2/m^2).

In this study, we consider vegetation cover to its full depth. In this case, $L = L_0$ and, for the phytocenosis albedo, the following formula is obtained from Eqs. (1) and (2):

$$A_f = 1 - \frac{KL\exp(-KL)}{1-a_T}. \tag{3}$$

On the basis of the generalization of observational data on the attenuation of the total radiant fluxes in different plant communities, semiempirical formulas are given in [17–19] that allow the attenuation to be calculated in relation to the leaf area and the Sun's altitude with consideration for the ratio between the fluxes of direct and scattered radiation over grass. The studies made by the authors of these works show that, despite the differences in geographical zones, plant heights, and in the values of the index of leaf area and its vertical distribution, radiation is attenuated according to the same law, and the total radiation transmission function $a_T(L, h_\Theta)$ can be approximated with the aid of the following semiempirical formula:

$$a_T(L, h_\Theta) = \frac{(S'/D)\exp[-c_1L/\sin h_\Theta] + a_D(L, h_\Theta)}{1 + S'/D} + c_2[\exp(c_1c_3L/\sin h_\Theta) - \exp(-c_1L/\sin h_\Theta)], \tag{4}$$

where h_Θ is the Sun's altitude; S'/D is the ratio of the flux density of direct radiation incident on the horizontal surface to the density of the flux density of scattered radiation; a_D is the function of transmission of radiation scattered by vegetation cover; and c_1, c_2, and c_3 are empirically determined constants (c_1 characterizes the effect of the geometric structure of vegetation cover on radiation transmission, c_2 determines the portion of radiation scattered by vegetation cover in the total radiation flux, and c_3 characterizes the intensity of radiation scattering by phytoelements and is related to their optical properties). An analysis of experimental data on light-flux transmission by grass phytocenoses within the PAR region [19] showed that, in this spectral region, the coefficient c_2 is small, and, thus, in Eq. (4), one can neglect the term describing the PAR scattered by phytoelements. More recent studies [7] showed that the function $a_T(L, h_\Theta)$ in the PAR region can be calculated from the formula

$$a_T(L, h_\Theta) = \frac{(S'/D)\exp[-G_L(h_\Theta)L/\sin h_\Theta] + a_D(L, h_\Theta)}{1 + S'/D}, \tag{5}$$

where $G_L(h_\Theta)$ is the function characterizing the spatial distribution of leaf area and its orientation.

Since we consider a grass phytocenosis to its full depth, we assume (without loss of generality) that its phytoelements are uniformly distributed; i.e., the same portion of their area is oriented in all directions. For this case, it is shown [18] that the function of the spatial orientation of leaves is constant:

$$G_L(h_\Theta) = 0.5$$

and the function of transmission of scattered radiation by vegetation cover has the form

$$a_D(L, h_\Theta) = 2E_3(0.5L),$$

where $E_3(0.5L)$ is the integral–exponential Gold function of the third order. Thus, for the case under consideration, Eq. (5) has the form

$$a_T(L, h_\Theta) = \frac{(S'/D)\exp[-0.5L/\sin h_\Theta] + 2E_3(0.5L)}{1 + S'/D}. \tag{6}$$

Table 1. Statistical characteristics of the LAI and albedo of grass cover for different ranges of its phytomass values

Characteristic	Ranges of phytomass variations, t/ha					
	0–2	2–4	4–6	6–8	8–10	10–12
	LAI, m^2/m^2					
Mean	1.6	3.9	4.5	3.4	4.2	3.5
Maximum	3.8	6.3	6.4	5.4	6.0	5.4
Minimum	0.4	1.7	2.3	1.2	2.4	1.8
Rms deviation	0.9	1.1	1.0	1.2	1.0	1.1
	Albedo					
Mean	0.126	0.189	0.223	0.234	0.202	0.215
Maximum	0.219	0.26	0.26	0.277	0.252	0.23
Minimum	0.05	0.119	0.173	0.172	0.120	0.208
Rms deviation	4.0	4.3	2.3	2.5	3.4	0.8

It should be noted that theoretical equation (3) for the albedo of vegetation cover within the PAR spectral region is obtained under the assumption that the structure of plant communities is ideal, which provides a maximum acceptance of carbon dioxide through a photosynthesis process and, as a consequence, a phytomass increase. However, the vegetation cover of the underlying surface has the property of self-regulation and does not always operate in an ideal mode; moreover, the measured values of the underlying-surface albedo are mainly the values of the integral albedo A. In order to fit the theoretical and experimental results, we propose to introduce the coefficient τ in Eq. (3):

$$A = 1 - \tau\frac{KL\exp(-KL)}{1-a_T}. \qquad (7)$$

A large body of data on the values of the LAI and phytomass of grass cover in different native zones has been published. The experimental data characterizing the dependence of the LAI of vegetation on its overground phytomass for both meadow grasses and a number of crops are used in this work. (These data were obtained during the combined studies of the photosynthetic productivity of plants in some regions of European Russia [20–30] and in Estonia [18, 31, 32], Moldavia, Tajikistan [18, 32], and Ukraine [33]. Table 1 gives the statistical characteristics of the LAI for different ranges of grass-cover phytomass values. The approximation of the dependence of the index of the leaf area of plant communities on their phytomass (Fig. 1) is obtained from experimental data. A statistical analysis (the Pearson criterion χ^2) showed that, at the significance level 0.01, the dependence used by us is in agreement with the experimental data. It follows from Fig. 1 that, for grass phytocenoses, if the phytomass increases to 4.8 t/ha, the LAI increases to about 4.9 m^2/m^2. A further increase of P to 7.1 t/ha results in a decrease of the LAI to 2.9 m^2/m^2. Then again, if the phytomass increases to 9.7 t/ha, the LAI increases to 4.7 m^2/m^2. A further increase in the phytomass is accompanied by a gradual decrease of the LAI. Note that, below, the phytomass of grass cover is taken to mean the phytomass of the plants above ground level.

A joint analysis of the values of the LAI and phytomass of grass communities shows that the land vegetation cover has its own biological regulatory properties, which are manifested in interrelated variations of the indicated parameters. These properties are related to the seasonal dynamics of grass-cover development. Plant communities consist of plant groups with different rhythms of development. Some of them reach the LAI maximum at a phytomass of 4.8 t/ha. A further increase of the phytomass of the plants of this group results in a decrease of their leaf area. For the other plant phytocenoses, the LAI reaches its maximum at a phytomass of 9.7 t/ha. These two groups of phytocenoses are, as a rule, separated in time [34]. The first group is characterized by nonzero values of the LAI during late April–early May, its maximum during the first half of June, and its further rapid decrease almost to zero by early July. The other group is characterized

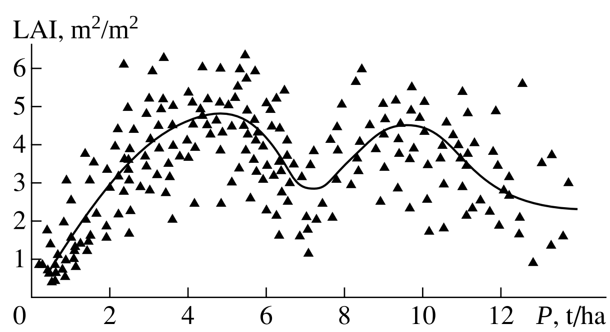

Fig. 1. Experimental data (triangles) according to [20–33] and the phytomass (P) dependence of the LAI (solid curve) for grass phytocenoses.

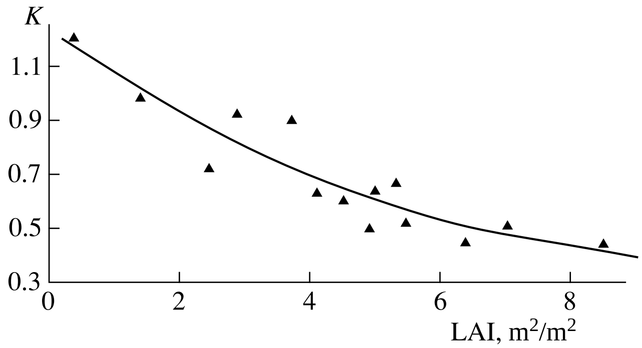

Fig. 2. Experimental data (triangles) according to [35] and the LAI dependence of the extinction coefficient (K) (solid curve) for grass cover.

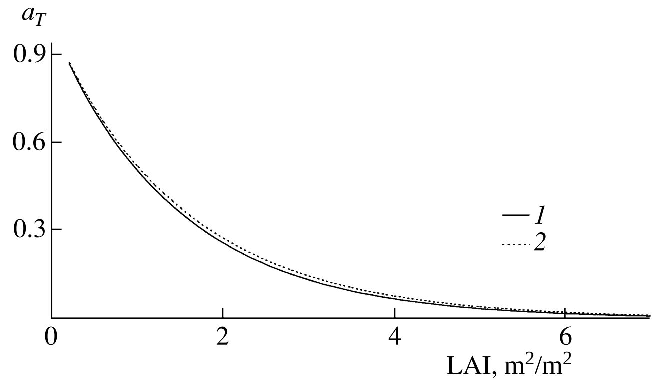

Fig. 3. Relation between light-flux attenuation (a_T) and the LAI for the grass communities of the (*1*) Leningrad and (*2*) Kursk regions.

by nonzero values of the LAI from late May, its maximum during late July–early August, with its further gradual decrease to zero by the second half of September. Thus, the temporal regulation of plants is observed during their vegetation period, which causes the spatiotemporal inhomogeneity of the underlying surface.

On the basis of empirical data, a general regularity is revealed in the LAI dependence of the extinction coefficient K [35]. The results of the experimental studies made by the authors of [35] showed that, in the plant communities with an underdeveloped leaf surface, the extinction coefficient is the highest; and the values of K decrease, while the plant community develops and a few layers of leaves appear. We plotted the LAI dependence of the extinction coefficient (Fig. 2) from the experimental data obtained for meadow phytocenoses. It follows from Fig. 2 that, for the communities under consideration, the values of K gradually decrease from 1.2 to 0.4, while the LAI increases from 0.2 to 9.0 m^2/m^2.

According to Eq. (6), at a certain altitude of the Sun and given geographical coordinates of the region under study, the function of light-flux attenuation $a_T(L, h_\Theta)$ within the grass-cover layer is a function of the LAI. From Eq. (7) and the resulting phytomass dependence of the LAI and relation between the extinction coefficient and the LAI, it follows that the albedo of the underlying surface covered by vegetation can be determined as a function of the phytomass of vegetation cover.

A semiempirical model proposed to calculate the integral albedo of the underlying surface in relation to the values of the phytomass of grass cover is identified on the basis of the experimental data obtained from both phytometric and actinometric studies simultaneously conducted by researchers from the Institute of Geography, Russian Academy of Sciences, in the Seim river basin (Kursk region) in 1973–1991 [11, 20, 24, 26, 30] and by those from the Main Geophysical Observatory near the town of Gatchina (Leningrad region) in 1965–1966 [21, 36, 37]. Since the reflectivity of the underlying surface varies depending on weather conditions and the Sun's altitude, we used the data obtained only during midday observations under clear skies to analyze the dependence of variations in the albedo of the underlying surface on its phytomass.

To calculate the function $a_T(L, h_\Theta)$ with the use of Eq. (6), it is necessary to determine the value of S'/D for the PAR spectral region. To this end, according to the methods used in [38], one can use the integral values of S' and D and multiply the ratio S'/D by 0.65. The ratios S'/D are calculated for the PAR region on the basis of the data (obtained by the network of actinometric stations [39]) on daytime variations in the mean intensity of direct radiation incident on the horizontal surface S' (W/m^2) and the mean intensity of scattered radiation D (W/m^2) under clear skies for the latitudes of the areas under consideration. During the vegetation period, which lasts from April to September in the Kursk region and from May to August in the Leningrad region, the resulting midday ratios S'/D vary within 3.1–3.8 and 3.5–4.1, respectively. The comparison between the LAI dependences of light-flux attenuation a_T for the regions under consideration at possible values of the ratio S'/D during the vegetation period showed that the values of this ratio can be replaced by its mean values over the vegetation period, which only slightly affects the accuracy. In this case, the error does not exceed 8%. Figure 3 shows the light-flux attenuation functions a_T (used in identifying the model) for the grass cover layer. It follows from Fig. 3 that the functions under consideration differ only slightly from one another (by no more than 0.02) within the interval of LAI variation from 1 to 4 m^2/m^2 and almost coincide at the rest of LAI values.

The coefficients τ were determined on the basis of the experimental data [11, 26, 36, 37] obtained for the grass ecosystems of the Kursk and Leningrad regions with the use of Eq. (7). The dependences of the coefficients τ on the values of the phytomass of the phytocenoses under consideration in the indicated regions (Fig. 4) were obtained with consideration for the dependence of the LAI for grass on the values of its phytomass. It follows from the dependences obtained that the coefficient τ for grass communities is unambiguously determined from the value of their phytomass.

RESULTS AND CONCLUSIONS

This study shows that the values of the coefficient τ for the regions under consideration are close to one another (Fig. 4). Therefore, one can assume that the phytomass dependence of the coefficient τ is a biological property of the grass cover of the underlying surface, is independent of a geographical location of investigations, and is common for underlying surfaces covered by both natural grass and crops. This assumption is supported by the fact that the obtained phytomass dependences of the coefficient τ (Fig. 4) are close to the phytomass dependence of the LAI (Fig. 1).

Variations in the integral albedo and phytomass of grass during different stages of its vegetation period can be analyzed on the basis of the results of the long-term simultaneous measurements of these characteristics conducted at the Kursk field station. Table 2 gives the statistical characteristics of the data obtained during the 1973–1988 field measurements. At the beginning of the vegetation period, grass has an insignificant phytomass (up to 0.25 t/ha) and a small leaf area (up to 0.15 m^2/m^2). Therefore, within this period, grass phytocenoses are characterized by a slight density, and their albedo is formed through reflection from grass and soil. During these days, the albedo has the smallest (over the whole vegetation period) values at a sufficiently high value of the variation coefficient (about 20%). During the first month of vegetation, the development of grass cover is characterized by an intensive increase of its phytomass (up to 3 t/ha). During this period, the density of grass cover increases, which results in a decrease of the coefficient of variation in its albedo to 7–8%. The grass albedo maximum is observed during the second half of June (on average, 0.23–0.24) at a phytomass of 5–6 t/ha with a low variation coefficient (8–10%). During this period, grain crops begin to ear and their leaf area begins to decrease. The phytomass of the grass ecosystems of the region under consideration reaches its maximum during the second half of July (8–9 t/ha) at a variation coefficient of 10%. During this period, the albedo value is about 0.22 at a variation coefficient of 4 to 10%.

Thus, the analysis of the experimental data showed that, for grass ecosystems, their albedo maximum does not correspond to their phytomass maximum. Table 1 gives the statistical characteristics of grass albedo obtained on the basis of the experimental data [11, 18, 26, 30, 33, 36, 37, 40] for different ranges of phytomass values.

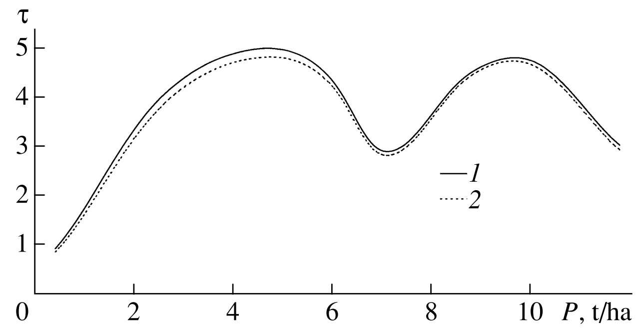

Fig. 4. Phytomass (P) dependence of the coefficient τ for the grass phytocenoses of the (*1*) Leningrad and (*2*) Kursk regions.

Figure 5 presents the albedo A–phytomass P relationship obtained from Eq. (7) with consideration for the dependences $L(P)$, $K(L)$, $a_T(L)$ and $\tau(P)$ (Figs. 1–4). The model curves lie within the experimentally measured values. It follows from the graphs presented that the albedo of grass-cover is unambiguously determined from the values of its phytomass. However, $A(P)$ is not a single-valued function.

At low values of the phytomass of grass communities, which correspond to their initial vegetation period, the grass-cover density is low. During this period, the soil type (chernozem and podzol for the Kursk and Leningrad regions, respectively) has a profound effect on the albedo value. This fact determines the initial values for the graphs presented in Fig. 5. A rapid increase in

Table 2. Variations in the albedo and phytomass of grass cover during its vegetation period: mean values ($\bar{x}$) and variation coefficient (v)

Month	Days elapsed from the start of vegetation period	Albedo		Phytomass	
		$\bar{x}_a$	v_a, %	$\bar{x}_p$, t/ha	v_p, %
IV	7	0.065	23.3	0.03	20.2
V	14	0.086	15.1	0.29	30.0
	21	0.116	14.4	1.03	28.2
	28	0.136	8.9	2.52	21.3
VI	35	0.184	8.4	3.97	12.9
	42	0.223	8.1	5.05	12.2
	49	0.235	7.4	5.72	8.4
	56	0.235	6.2	6.16	10.3
VII	63	0.232	5.8	6.44	12.1
	70	0.231	4.7	6.89	14.3
	75	0.222	4.0	7.55	15.2
	80	0.218	9.2	8.33	10.1
	85	0.218	10.4	8.99	10.1
VIII	92	0.187	12.1	8.97	8.3
	101	0.142	10.2	8.64	5.4

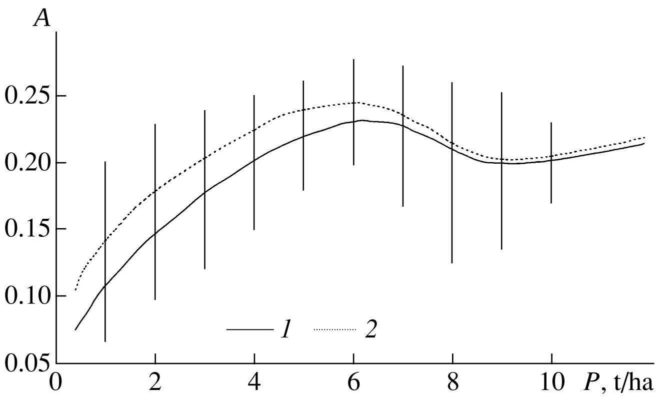

Fig. 5. Phytomass dependence of albedo (*A*) for grass cover: model calculations for the (*1*) Leningrad and (*2*) Kursk regions; the vertical lines show the range of variations in the experimental values of *A* at a given value of *P*.

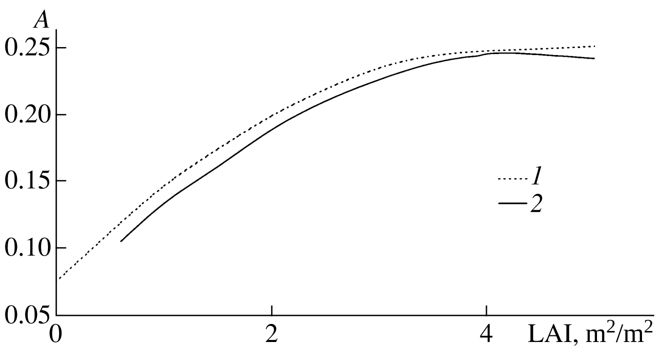

Fig. 6. LAI dependence of albedo (*A*) for the Kursk grasses: (*1*) experimental data according to [11] and (*2*) model calculations.

the albedo values (from 0.11–0.14 to 0.23–0.24) is characteristic of the interval of *P* variations between 1 and 6 t/ha. Such a rapid increase of the albedo values within this range of phytomass variations is due to an increase in the mass of grass cover and changes in its architectonics. If the phytomass values increase from 5.7 to 6.5 t/ha, the albedo of grass cover remains almost unchanged (about 0.23 for the Kursk region and 0.24 for the Leningrad region). High values for the phytomass of plants and for their leaf area are characteristic of this period. During this period, the effect of soil on the reflectivity of the soil–grass system is decreased, and the reflectivity of leaves remains almost unchanged. This implies that the grass cover has formed its microclimate and structure. It should be noted that, at a production of 6 t/ha, the narrowest range of variations is observed in the experimentally obtained albedo values. With a further increase of the phytomass of grasses, the area of leaves decreases and the portion of generative organs (ears, flowers) in the vegetation structure increases. During this period, an albedo decrease is observed. At the end of the vegetation period, grain crops change their color from green to yellow and even to whitish; in this case, the albedo is slightly increased [41], which is manifested by the graph of the dependence $A(P)$ obtained with the model. It should be noted that, within the range of phytomass values from 6 to 11 t/ha, the albedo varies within 0.2–0.24. Thus, while the phytomass values are changed twofold within this range, the albedo value is changed only by 0.04.

To verify the functional dependence, we used the phytomass and albedo data obtained for spoil grounds in the Kursk region in 1981–1985 [26]. The values of the phytomass of grass covering the four-year-old spoil grounds (the initial stage) are not high (about 2.1 t/ha). The phytomass of the grass covering the ten-year-old spoil grounds increases to 2.3 t/ha. A further increase in the phytomass to 3.2 t/ha is attributed to the initial stage of formation of meadows on the spoil grounds. The reflectivity of grass covering the spoil grounds of different age is different. Albedo values 0.14–0.17 are characteristic of the soil grounds covered with grass at its initial stage of development. Albedo values of about 0.18–0.2 are characteristic of ten-year-old phytocenoses. A meadow stage is characterized by albedo values ranging from 0.19 to 0.22. The model calculations of the albedo values for the phytomasses characteristic of the three stages under consideration yield the following results: 0.17, 0.19, and 0.21, respectively. A comparison between the albedo values obtained experimentally and theoretically on the basis of Eq. (7) shows their satisfactory agreement.

The model under consideration allows one to determine the LAI dependence of albedo. The LAI dependence of *A* (Fig. 6) (obtained with the model) is in good agreement with a similar dependence obtained on the basis of the generalized experimental data [11].

Thus, on the basis of the available experimental data and the results of theoretical studies, it is possible to establish the dependence of integral-albedo variations on the phytomass of natural grass communities. This dependence allows one to roughly estimate the integral albedo of grass from a specified value of its phytomass.

This study allows one to conclude that, during the vegetation period, the albedo of grass ecosystems varies, on average, from 0.09 to 0.25. This range of albedo values corresponds to phytomass values varying from 0.04 to 12 t/ha. It follows from the results obtained that the problem of determining the phytomass from albedo values has an unambiguous solution only for albedo values of 0.09 to 0.2 at phytomass values reaching 4 t/ha, which corresponds to the initial stage of grass development. For higher albedo values, which correspond to phytomass values varying from 4 to 12 t/ha, this problem has an ambiguous solution. It should be noted that a rather wide range of variations in the LAI (2.1–5.0 m^2/m^2) corresponds to the indicated range of albedo variations (0.2–0.25).

Despite its approximate character and a number of simplifying assumptions, the model under consideration allows one to correctly describe the response of the optical properties of grass ecosystems to variations in their productivity and does not contradict experimental data.

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